

Solution to KenKen (12 × 12)

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Preface

This solution presents one of the many ways to solve this KenKen. Per usual, standard KenKen rules apply, and in the following steps visual aid will be provided in order to help depict where certain numbers go within the KenKen. The KenKen itself will be centered inside a grid labeled horizontally (A-L) and vertically (1-12) in order to indicate specifically which square (referred to as a cell throughout this solution) a number will be placed in. The naming system for these squares will work like a chess board: Each letter will refer to the given column the square is in and each number will refer to the given row the square is in, so where the given column and row intersect is where you will find your square.

1 Fill in the givens

We can start by filling in the single cells that already have a given number, and we get this grid:

	A	B	C	D	E	F	G	H	I	J	K	L
1				630×	12	8×		360×	81×			44×
2	60×	10×		15+	44×						13+	12
3				3×			16+		490×	486×	12	44×
4		128×	60×	12	50×		165×	3+				
5		11		4050×					48×	12		14×
6	12		3+			14×		220×		21×	48×	80×
7	77×	648×		1-			12					
8						132×		20×	12		5×	18×
9		12		70×		3÷	9×	72×		100×		
10	11+		110×		22+	12			8×			105×
11			12			378×	28×	11			160×	
12			60×			11		12				19+

You will notice that all of the 12s have been given, in addition to some 11s. We will keep this in mind later when solving the rest of the puzzle as we can no longer count 12 as a possibility in any future cell.

2 Determine possibilities that are definite for operations on two cells

- Set: Cells F1 and F2 which multiply to 8. The possible solutions are 1×8 and 2×4 . We will determine which of these is the correct solution later.
- Set: Cells A2 and A3 which multiply to 60. The only possible solutions are 6×10 and 5×12 . However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×10 .
- Set: Cells B2 and B3 which multiply to 10. The only possible solutions are 1×10 and 2×5 . We will determine which of these is the correct solution later.
- Set: Cells C3 and D3 which multiply to 3. The only possible solution is 1×3 .
- Set: Cells E2 and E3 which multiply to 44. The only possible solution is 4×11 .
- Set: Cells L3 and L4 which multiply to 44. The only possible solution is 4×11 .
- Set: Cells C4 and C5 which multiply to 60. The only possible solutions are 6×10 and 5×12 . However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×10 .
- Set: Cells H4 and H5 which add to 3. The only possible solution is $1 + 2$.
- Set: Cells I5 and I6 which multiply to 48. The only possible solutions are 4×12 and 6×8 . However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×8 .
- Set: Cells K5 and L5 which multiply to 14. The only possible solution is 2×7 .
- Set: Cells B6 and C6 which add to 3. The only possible solution is $1 + 2$.
- Set: Cells F6 and F7 which multiply to 14. The only possible solution is 2×7 .
- Set: Cells J6 and J7 which multiply to 21. The only possible solution is 3×7 .
- Set: Cells K6 and K7 which multiply to 48. The only possible solutions are 4×12 and 6×8 . However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 6×8 .
- Set: Cells L6 and L7 which multiply to 80. The only possible solution is 8×10 .
- Set: Cells G8 and H8 which multiply to 20. The only possible solutions are 2×10 or 4×5 . We will determine which of these is the correct solution later.
- Set: Cells L8 and L9 which multiply to 18. The only possible solutions are 2×9 and 3×6 . We will determine which of these is the correct solution later.

- Set: Cells C9 and D9 which multiply to 70. The only possible solution is 7×10 .
- Set: Cells E9 and F9 which divide to 3. The only possible solutions are $12 \div 4$, $9 \div 3$, $6 \div 2$, or $3 \div 1$. We know that all the 12s in this puzzle have already been put in place, so this leaves us with the last three solutions as possible combinations.
- Set: Cells G9 and G10 which multiply to 9. The two possible solutions are 9×1 or 3×3 . However, 3×3 would not work because we cannot have more than one 3 in a singular column, so we know that these two cells will contain 9 and 1.
- Set: Cells H9 and H10 which multiply to 72. The only possible solutions are 6×12 and 8×9 . However, we know that all the 12s in this puzzle have already been put in place, so the only possible solution is 8×9 .
- Set: Cells B10 and C10 which multiply to 110. The only possible solution is 10×11 .
- Set: Cells G11 and G12 which multiply to 28. The only possible solution is 4×7 .
- Set: Cells K12 and L12 which add to 19. The only possible solutions are $7+12$, $8+11$, or $9+10$. We know that we already have an 11 and 12 in this row, so the only possibility is $9+10$.

The rest of the cells with sets of two have more possibilities, so we can leave them for later.

3 Determine possibilities that are definite for operations on three cells

- Set: Cells G1, G2, and H1 which all multiply to 360. The prime factorization of 360 is $2^3 \times 3^2 \times 5$. The only possible combinations from this are $4 \times 9 \times 10$, $5 \times 8 \times 9$, or $6 \times 6 \times 10$. We will determine which of these is the correct solution later.
- Set: Cells I1, I2, and H2 which multiply to 81. The prime factorization of 81 is 3^4 . The only two ways that we can group these 3s are $3 \times 3 \times 9$ ($I1=3$, $I2=9$, $H2=3$) or $1 \times 9 \times 9$ ($I1=9$, $I2=1$, $H2=9$). We will determine which of these is the correct solution later.
- Set: Cells J1, K1, and L1 which all multiply to 44. The prime factorization of 44 is $2^2 \times 11$. The only way we can group these numbers so that there is no repetition is $1 \times 4 \times 11$.
- Set: Cells H3, I3, and I4 which all multiply to 490. The prime factorization of 490 is $2 \times 5 \times 7 \times 7$, so in order to fit this into 3 cells, we must have $10 \times 7 \times 7$. (None of the other solutions of $14 \times 5 \times 7$, $35 \times 2 \times 7$, or $49 \times 2 \times 5$ have all three numbers between 1 and 12.) In addition, since we know that we cannot repeat numbers within columns or rows, we must place a 7 in H3, a 10 in I3, and a 7 in I4.
- Set: Cells J3, J4, and K4 which all multiply to 486. The prime factorization of 486 is 2×3^5 . The only way we can group these numbers such that all three are between 1 and 12 is $6 \times 9 \times 9$. In addition, since we cannot repeat numbers within columns or rows, we must place a 9 in J3, a 6 in J4, and a 9 in K4.
- Set: Cells A4, A5, and B4 which all multiply to 128. The prime factorization of 128 is 2^7 or $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$. The only two ways that we can group these 2s to create three numbers is $2 \times 8 \times 8$ and $4 \times 4 \times 8$. We will determine which is the correct solution later.

- Set: Cells E4, E5, and F5 which all multiply to 50. The prime factorization of 50 is $2 \times 5 \times 5$. The only ways we can group these numbers are $2 \times 5 \times 5$ or $1 \times 5 \times 10$. We will determine which is the correct solution later.
- Set: Cells G4, G5, and G6 which all multiply to 165. The prime factorization of 165 is $3 \times 5 \times 11$, which is the only possible solution.
- Set: Cells H6, H7, and I7 which all multiply to 220. The prime factorization of 220 is $2 \times 2 \times 5 \times 11$. The only possible ways to group these numbers are $2 \times 10 \times 11$ or $4 \times 5 \times 11$. We will determine which is the correct solution later.
- Set: Cells A7, A8, and A9 which all multiply to 77. The prime factorization of 77 is 7×11 , so the only possible solution is $1 \times 7 \times 11$.
- Set: Cells B7, B8, and C8 which all multiply to 648. The prime factorization of 648 is $2^3 \times 3^4$ and the only way we can group these numbers such that they are between 1 and 11 (not 12 since every row and column already has a 12) is $8 \times 9 \times 9$. In addition, since we know that we cannot repeat numbers within columns or rows, we must place a 9 in B7, an 8 in B8, and a 9 in C8.
- Set: Cells D8, E8, and F8 which all multiply to 132. The prime factorization of 132 is $2 \times 2 \times 3 \times 11$, and the only ways we can group these numbers such that they are between 1 and 11 (not 12 since every row and column already has a 12) are $2 \times 6 \times 11$ or $3 \times 4 \times 11$. We will determine which is the correct solution later.
- Set: Cells J8, K8, and K9 which all multiply to 5. Since 5 is prime, the only solution is $5 \times 1 \times 1$. In addition, since we cannot repeat numbers in columns or rows, there must be a 1 in J8, a 5 in K8, and a 1 in K9.
 - We also can conclude that there must be a 9 in G9 and a 1 in G10 (see Step 2 for reference) because there cannot be another 1 in Row 9.
- Set: Cells I9, J9, and J10 which all multiply to 100. The prime factorization of 100 is $2 \times 2 \times 5 \times 5$, and the only possible groupings of these numbers are $1 \times 10 \times 10$, $2 \times 5 \times 10$, or $4 \times 5 \times 5$. We will determine which is the correct solution later.
- Set: Cells I10, I11, and I12 which all multiply to 8. The prime factorization of 8 is 2^3 . However, we know that we cannot repeat numbers within columns, so the only possible grouping of these 2s is $1 \times 2 \times 4$.
- Set: Cells K10, L10, and L11 which all multiply to 105. The prime factorization of 105 is $3 \times 5 \times 7$, and this is the only possible grouping of these numbers, since they are all prime.
- Set: Cells A12, B12, and C12 which all multiply to 60. The prime factorization of 60 is $2 \times 2 \times 3 \times 5$, and the only possible groupings of these numbers are $2 \times 6 \times 5$, $2 \times 3 \times 10$ or $3 \times 4 \times 5$. We will determine which is the correct solution later.
- Set: Cells J11, J12, and K11 which all multiply to 160. The prime factorization of 160 is $2^5 \times 5$, and the only possible ways to group these numbers are $2 \times 8 \times 10$, $4 \times 4 \times 10$, $4 \times 5 \times 8$. We will determine which is the correct solution later.

The rest of the cells with sets of three have more possibilities, so we can leave them for later.

4 Determine possibilities that are definite for operations on four cells

- Set: Cells A1, B1, C1, and D1 which all multiply to 630. The prime factorization of 630 is $2 \times 3 \times 3 \times 5 \times 7$, so the only possible solutions are $1 \times 7 \times 9 \times 10$, $2 \times 5 \times 7 \times 9$, and $3 \times 5 \times 6 \times 7$.
- Set: Cells D5, D6, E6, and E7 which all multiply to 4050. The prime factorization of 4050 is $2 \times 3 \times 3 \times 3 \times 3 \times 5 \times 5$. We have to group this into $5 \times 9 \times 9 \times 10$, or else we have too many numbers or numbers that are too large.
- Set: Cells D12, E11, E12, and F11 which all multiply to 378. The prime factorization of 378 is $2 \times 3 \times 3 \times 3 \times 7$. The only possible groupings of these numbers are $1 \times 6 \times 7 \times 9$ or $2 \times 3 \times 7 \times 9$. We will determine which is the correct solution later.

After Steps 2, 3, and 4, you should have the following grid:

	A	B	C	D	E	F	G	H	I	J	K	L
1	$2 \times 5 \times 7 \times 9$	$3 \times 5 \times 6 \times 7$	$1 \times 7 \times 9 \times 10$	$630 \times$	12	$8 \times$	$4 \times 9 \times 10$ $5 \times 8 \times 9$ $6 \times 6 \times 10$	$360 \times$	$81 \times$	$1 \times 4 \times 11$		$44 \times$
2	$60 \times$	$10 \times$		$15 \times$	$44 \times$	1×8 or 2×4			$1 \times 1 \times 9$ $3 \times 3 \times 9$		$13 \times$	12
3	6×10	1×10 or 2×5	1×3	$3 \times$	4×11		$16 \times$	7	10	9	12	$44 \times$
4	$2 \times 8 \times 8$ $4 \times 4 \times 8$	$128 \times$	$60 \times$	12	$50 \times$		$165 \times$	$3 \times$	7	6	9	4×11
5		11	6×10	$4050 \times$	$1 \times 5 \times 10$ $2 \times 5 \times 5$		$3 \times 5 \times 11$	1×2	$48 \times$	12	2×7	$14 \times$
6	12	1×2	$3 \times$	$5 \times 9 \times 9 \times 10$		$14 \times$		$220 \times$	6×8	$21 \times$	$48 \times$	$80 \times$
7	$77 \times$	9		$1 \times$		2×7	12	$2 \times 10 \times 11$ $4 \times 5 \times 11$		3×7	6×8	8×10
8	$1 \times 7 \times 11$	8	9	$2 \times 6 \times 11$ $3 \times 4 \times 11$		$132 \times$	2×10 or 4×5	$20 \times$	12	1	5	$18 \times$
9		12	7×10	$70 \times$	$9 \div 3$ $6 \div 2$ $3 \div 1$	$3 \div$	9	$72 \times$	$1 \times 10 \times 10$ $2 \times 5 \times 10$ $4 \times 5 \times 5$	$100 \times$	1	2×9 or 3×6
10	$11 \times$	10×11	$110 \times$		$22 \times$	12	1	8×9	$8 \times$		$3 \times 5 \times 7$	$105 \times$
11			12			$378 \times$	$28 \times$	11	$1 \times 2 \times 4$	$2 \times 8 \times 10$ $4 \times 4 \times 10$ $4 \times 5 \times 8$	$160 \times$	
12	$2 \times 5 \times 6$ $2 \times 3 \times 10$ $3 \times 4 \times 5$		$60 \times$	$1 \times 6 \times 7 \times 9$ $2 \times 3 \times 7 \times 9$		11	4×7	12			9×10	$19 \times$

5 Take the definite solutions to cells to eliminate possible solutions for other cells, as well as fill some cells based on the same definite solutions

- We can eliminate the solution $1 \times 7 \times 9 \times 10$ for Cells A1, B1, C1, and D1. In the same row (Cells J1, K1, L1) there is already going to be a 1 within one of the cells as a definite solution, so the 1 for this row cannot come from Cells A1, B1, C1, and D1.

- We can eliminate the solutions $4 \times 9 \times 10$ and $5 \times 8 \times 9$ for cells G1, G2, and H1. In Columns G and H there are already going to be definite 9s in each column (G9 and H9 or H10). From here, we are left with the solution $6 \times 6 \times 10$, and we know that we cannot repeat numbers in columns or rows. There must be a 10 in G1, a 6 in G2, and a 6 in H1.
 - From here we can also eliminate $3 \times 5 \times 6 \times 7$ as a solution from Cells A1, B1, C1, and D1 because there is already a 6 in the first row. This leaves us with only the solution of $2 \times 5 \times 7 \times 9$ to fill these cells.
- We must put a 1 in L1. Columns J and K already have 1s in them (J8 and K9), so L1 is the only cell to place the 1 for the solution of J1, K1, and L1.
- We must place a 10 in A2 and a 6 in A3. Row 2 already has a 6 in it (G2), so we have to place the 6 of the solution for A2 and A3 in A3. This leaves us with only A2 to place the 10.
- We can eliminate the solution 1×10 for Cells B2 and B3. Both Rows 2 and 3 already have 10s in them (A2 and I3), so 10 for these rows is not going to come from B2 or B3. This leaves us with the solution of 2×5 for Cells B2 and B3.
 - From here we also know that we must place a 1 in B6 and a 2 in C6. Column B already has a 2 in it (B2 or B3), so we have to place the 2 of the solution for B6 and C6 in C6. This leaves us with only B6 to place the 1.
- We must place a 10 in C4 and a 6 in C5. Row 4 already has a 6 in it (J4), so we have to place the 6 of the solution for C4 and C5 in C5. This leaves us with only C4 to place the 10.
- We must place a 2 in H4 and a 1 in H5. Row 5 already has a 2 in it (K5 or L5), so we have to place the 2 of the solution for H4 and H5 in H4. This leaves us with only H5 to place the 10.
 - From here, we can also eliminate $2 \times 8 \times 8$ as a solution for Cells A4, A5, and B4 because in this solution we would have to place a 2 in Row 4, however there is already a 2 in Row 4 (H4). This leaves us with the solution $4 \times 4 \times 8$, and we know that we cannot repeat numbers in columns or rows. There must be an 8 in A4, a 4 in A5, and a 4 in B4.
 - * From here, we also know that we must place a 4 in L3 and an 11 in L4. Row 4 already has a 4 in it (B4), so we have to place the 4 of the solution for L3 and L4 in L3. This leaves us with only L4 to place the 11.
 - * From here, we also know that we must place a 4 in E2 and an 11 in E3. Row 3 already has a 4 in it (L3), so we have to place the 4 of the solution for E2 and E3 in E2. This leaves us with only E3 to place the 11.
 - From here, we can also eliminate $2 \times 5 \times 5$ as a solution for cells E4, E5, and F5, because there is already going to be a 2 in Rows 4 and 5 (H4 and K5 or L5). This leaves us with the solution $1 \times 5 \times 10$. Since there already is a 10 in Row 4 and a 1 in Row 5, we must put the 10 in Row 5 (E5 or F5) and the 1 in Row 4 (E4).
 - From here, we can also eliminate $1 \times 1 \times 9$ as a solution for Cells H2, I1, and I2 because there is already going to be a 1 in Column H (H5), the column in which a 1 would have gone in if we chose the solution set $1 \times 1 \times 9$. This leaves us with the solution $3 \times 3 \times 9$. We know that we cannot repeat numbers in columns or rows. There must be a 3 in H2, a 3 in I1, and a 9 in I2.

- We must place an 8 in I5 and a 6 in I6. Row 5 already has a 6 in it (C5), so we have to place the 6 of the solution for I5 and I6 in I6. This leaves us with only I5 to place the 8.
- We must place an 8 in K6 and a 6 in K7. Row 6 already has a 6 in it (I6), so we have to place the 6 of the solution for K6 and K7 in K7. This leaves us with only K6 to place the 8.
- We must place a 10 in L6 and an 8 in L7. Row 6 already has an 8 in it (K6), so we have to place the 8 of the solution for L6 and L7 in L7. This leaves us with only L6 to place the 10.
 - From here, we also know that we must place a 10 in E7. The set of Cells D5, D6, E6, and E7 which all multiply to 4050 have the solution $5 \times 9 \times 9 \times 10$, and we know the 10 cannot be placed in Rows 5 or 6 because there are already 10s in these rows (E5/F5 and L6). We can also place the 5 and 9s as we know that we cannot repeat numbers in columns or rows. There must be a 9 in D5, a 5 in D6, and a 9 in E6.
 - * From here, we must place a 5 in E5 and a 10 in F5. Column E already has a 10 in it (E7), so we have to place the 10 of the solution for E5 and F5 in F5. This leaves us with only E5 to place the 5.
 - * From here we can also place an 11 in G6 since the solution for Cells G4, G5, and G6 must be $3 \times 5 \times 11$, and there is already a 3 and 5 in Row 6 (J6 for 3 and D6 for 5), so the only number that can be placed in G6 from this solution set is 11. We must also place a 5 in G4 and a 3 in G5. Row 5 already has a 5 in it (E5), so we have to place the 5 of the solution for G4 and G5 in G4. This leaves us with only G5 to place the 3.
- We can eliminate the solution $2 \times 10 \times 11$ for cells H6, H7, and I7. Both Columns H and I already have 2s in them (H4 and I10/I11/I12). This means that the 2 for these columns is not going to come from H6 or H7 or I7. We are left with the solution $4 \times 5 \times 11$, and we can put the 11 in I7 since Column H already has an 11 in it (H11). We must place a 4 in H6 and a 5 in H7. Row 6 already has a 5 in it (D6), so we have to place the 5 of the solution for H6 and H7 in H7. This leaves us with only H6 to place the 4.
- We must place a 7 in F6 and a 2 in F7. Row 6 already has a 2 in it (C6), so we have to place the 2 of the solution for F6 and F7 in F7. This leaves us with only F6 to place the 7.
 - From here, we also know that we must place a 3 in J6 and a 7 in J7. Row 6 already has a 7 in it (F6), so we have to place the 7 of the solution for J6 and J7 in J7. This leaves us with only J6 to place the 3.
 - From here, we can also eliminate the solution 2×4 for cells F1 and F2. Column F already has a 2 in it (F7), so the 2 for this column cannot come from F1 or F2. This leaves us with the solution of 1×8 for cells F1 and F2. We also know that we must place the 1 in F2 and the 8 in F1. Row 1 already has a 1 in it (L1), so we have to place the 1 of the solution for F1 and F2 in F2. This leaves us with only F1 to place the 8.
- We must place a 1 in A7 since the solution for Cells A7, A8, and A9 must be $1 \times 7 \times 11$, and there is already a 7 and 11 in Row 7 (J7 for 7 and I7 for 11), so the only number that can be placed in A7 from this solution set is 1. We must also place a 7 in A8 and an 11 in A9. Row 9 already has a 7 in it (C9/D9), so we have to place the 7 of the solution for A8 and A9 in A8. This leaves us with only A9 to place the 11.

- We can eliminate the solution 4×5 for Cells G8 and H8. Row 8 already has a 5 in it (K8), so 5 for these rows cannot come from G8 or H8. This leaves us with the solution of 2×10 for Cells G8 and H8. We also know that we must place the 2 in G8 and the 10 in H8. Column H already has a 2 in it (H4), so we have to place the 2 of the solution for G8 and H8 in G8. This leaves us with only H8 to place the 10.
 - From here, we can also eliminate $2 \times 6 \times 11$ as a solution for Cells D8, E8, and F8, because there is already going to be a 2 in Row 8 (G8). This leaves us with the solution $3 \times 4 \times 11$. Since there already is an 11 in Column F (F12) and Column E (E2 or E3), we must put the 11 in D8. There already is a 4 in Column E (E2 or E3), so we must put the 4 in F8. This leaves us with only E8 to place the 3.
 - From here we can also eliminate 2×9 as a solution for cells L8 and L9 because there is already going to be a 2 and 9 in Row 8 (G8 for 2 and C8 for 9). This leaves us with the solution 3×6 . Since there already is a 3 in Row 8 (D8/E8/F8), we have to place the 3 of the solution for L8 and L9 in L9. This leaves us with only L8 to place the 6.
- We must place a 10 in D9 and a 7 in C9. Column C already has a 10 in it (C4), so we have to place the 10 of the solution for C9 and D9 in D9. This leaves us with only C9 to place the 7.
- We can eliminate the solutions $9 \div 3$ and $3 \div 1$ for Cells E9 and F9. Row 9 already has a 9 in it (G9), a 3 in it (L9), and a 1 in it (K9), so the 1, 3, and 9 for this row cannot come from E9 or F9. This leaves us with the solution of $6 \div 2$ for Cells E9 and F9. We must place 6 in F9 and 2 in E9. Column F already has a 2 in it (F7), so we have to place the 2 of the solution for E9 and F9 in E9. This leaves us with only F9 to place the 6.
- We must place a 9 in H10 and an 8 in H9. Row 9 already has a 9 in it (G9), so we have to place the 9 of the solution for H9 and H10 in H10. This leaves us with only H9 to place the 8.
- We can eliminate the solutions $1 \times 10 \times 10$ and $2 \times 5 \times 10$ for cells I9, J9, and J10. Row 9 already has a 10 in it (D9), a 1 in it (K9), and a 2 in it (E9), so the 1, 2, and 10 for this row cannot come from I9 or J9. This leaves us with the solution of $4 \times 5 \times 5$ for cells I9, J9, and J10. Since we cannot repeat numbers in columns or rows, there must be a 5 in I9, a 4 in J9, and a 5 in J10.
 - From here, we also know that we must place an 11 in J1 and a 4 in K1. Column J already has a 4 in it (J9), so we have to place the 4 of the solution for J1 and K1 in K1. This leaves us with only J1 to place the 11.
- We must place a 10 in B10 and an 11 in C10. Column B already has an 11 in it (B5), and Column C already has a 10 in it (C4), so we have to place the 10 and 11 in the opposite columns of the existing 10 and 11s, and yielding a 10 in B10 and an 11 in C10.
- We must place a 3 in K10, a 5 in L11, and a 7 L10. Column L already has 3 in it, so the 3 has to go into Column K and the only cell in this set it can occupy is K10. There is also already a 5 in Row 10, so the 5 has to go into Row 11, and the only cell in this set it can occupy is L11. Finally, this leaves us with only cell L10 to place the 7.
 - From here, we also know that we must place 7 in K5 and a 2 in L5. Column L already has a 7 in it (L10), so we have to place the 7 of the solution for K5 and L5 in K5. This leaves us with only L5 to place the 2.

- We must place a 10 in K12 and a 9 in L12. Column L already has a 10 in it (L6), so we have to place the 10 in K12. This leaves us with only L12 to place the 9.
 - From here, we can also eliminate $2 \times 3 \times 10$ as a solution for Cells A12, B12, and C12 because there is already going to be a 10 in Row 12 (K12). This leaves us with the solutions $2 \times 5 \times 6$ and $3 \times 4 \times 5$.
 - From here, we can also place a 9 in F11 because the possible solution for Cells D12, E11, E12, and F11 all have to contain a 9 within them; however, Row 12 already has a 9 in it (L12), and Column E already has a 9 in it (E6), so the only place for the 9 of this set to go in is F11. We can also eliminate the solution $2 \times 3 \times 7 \times 9$ because if we used this solution, we would have to place the 2 in D12 (Column E already has a 2 in E9) but also the 3 in D12 (Column E also already has a 3 in E8), and it is not possible to put two numbers within one cell. This leaves us with the solution $1 \times 6 \times 7 \times 9$. We can also place a 1 in D12 because Column E already has a 1 in it, so the only cell within the set that we can place the 1 in is D12.
 - * From here, we also know that we must place 1 in C3 and a 3 in D3. Column D already has a 1 in it (D12), so we have to place the 1 of the solution for C3 and D3 in C3. This leaves us with only D3 to place the 3.
- We can eliminate the solutions $4 \times 4 \times 10$ and $4 \times 5 \times 8$ for cells J11, J12, and K11. Columns J and K already have 4s in them (J9 and K1), so the 4 for these columns cannot come from J11 or J12 or K11. This leaves us with the solution of $2 \times 8 \times 10$ for Cells J11, J12, and K11. We can put the 8 in J12 because we know that within the possible solutions to cells in Row 12, J12 is the only place where we can get an 8 from. From here, we can also place 10 in J11 because Column K already has a 10 in it (K12) and this leaves us with only Cell K11 to place the 2.

After Step 5, you should obtain the following grid:

	A	B	C	D	E	F	G	H	I	J	K	L
1	2×5×7×9			630×	12	8	10	6	3	11	4	1
2	10	10×		15+	4	1	6	3	9		13+	12
3	6	2×5	1	3	11		16+	7	10	9	12	4
4	8	4	10	12	1		5	2	7	6	9	11
5	4	11	6	9	5	10	3	1	8	12	7	2
6	12	1	2	5	9	7	11	4	6	3	8	10
7	1	9		1-	10	2	12	5	11	7	6	8
8	7	8	9	11	3	4	2	10	12	1	5	6
9	11	12	7	10	2	6	9	8	1	5	1	3
10	11+	10	11	6	8	12	1	9	8×	1	3	7
11			12	8		9	28×	11	1×2×4	10	2	5
12	2×5×6 3×4×5		60×	1		11	4×7	12		8	10	9

6 Fill in the remaining blanks to finish the puzzle

- Starting with Cells A1, B1, C1, and D1, we know that the solution for this set is going to be $2 \times 5 \times 7 \times 9$. We know that we must place 7 in B1 because Column B already has a 2, 5, and 9 (B2, B3, B7). Next, we must place 9 in A1 because Columns B, C, and D already have a 9 in them (B7, C8, D5). Finally, we must place 5 in C1 and 2 in D1. Column C already has a 2 in it (C6), so we have to place the 2 of the solution for C1 and D1 in D1. This leaves us with only C1 to place the 5.
- In Column G, we notice that the only missing number is 8, so we must place an 8 in G3. From here, we can also notice that the only missing numbers in column F are 3 and 5. We must place a 5 in F3 and a 3 in F4. Row 3 already has a 3 in it (C3 or D3), so we have to place the 3 of the solution for F3 and F4 in F4. This leaves us with only F3 to place the 5. We now know that the solution set for Cells F3, F4, and G3 is $3+5+8$.
 - From here, we also know that we must place a 5 in B2 and a 2 in B3. Row 3 already has a 5 in it (F3), so we have to place the 5 of the solution for B2 and B3 in B2. This leaves us with only B3 to place the 2.
 - From here, we can notice that the missing numbers in Row 2 are 2, 8, 7, and 11 and the missing numbers in Columns J and K are 2 and 11, respectively. This means that we

must place an 8 in C2, a 7 in D2, a 2 in J2, and an 11 in K2. We know that the missing numbers in Columns J and K will be filled in, so the only numbers left in Row 2 will be 7 and 8. However, there is already a 7 in Column C (C9), so we can't place the 7 in C2, and we have to place it in D2. This leaves us with only cell C2 to place the 8.

- In Row 7 we can notice that the only missing numbers are 3 and 4. Therefore, know that the solution for Cells C7 and D7 will be 4-3. We must place the 3 in C7 and the 4 in D7. Column D already has a 3 in it (D3), so we have to place the 3 of the solution for C7 and D7 in C7. This leaves us with only D7 to place the 4.
- We must place a 1 in I11. Rows 10 and 12 already have 1s in them (G10 and D12), so we have to place the 1 in Row 11.
- In Columns D and E, we notice that the 8s for these columns are missing. The only set of cells where the 8s can come from are Cells D10, D11, and E10 which add up to 22. We know that we cannot repeat numbers in columns or rows. We must place the 8s in separate rows, so we will place an 8 in D11 and an 8 in E10. We know that Cells D10, D11, and E10 will all have to add up to 22, and the only number that will allow these cells to add up to 22 is 6, so we must place a 6 in D10.
- In Column C, we notice that the only missing number is 4, so we must place a 4 in C12. This means that the solution set for Cells A12, B12, and C12 is $3 \times 4 \times 5$. We must place the 5 in A12 and the 3 in B12. Column B already has a 5 in it (B2), so we have to place the 5 of the solution for A12 and B12 in A12. This leaves us with only B12 to place the 3.
- In Column B, we notice that the only missing number is 6, so we must place a 6 in B11. The remaining numbers in the solution set for Cells A10, A11, and B11 have to be 2 and 3 (the only remaining numbers in Column A). We must place the 2 in A10 and the 3 in A11. Row 10 already has a 3 in it (K10), so we have to place the 3 of the solution for A10 and A11 in A11. This leaves us with only A10 to place the 2.
- We must place a 6 in E11 and a 7 in E12. Row 11 already has a 6 in it (B11), so we have to place the 6 of the solution for E11 and E12 in E11. This leaves us with only E12 to place the 7.
- We must place a 4 in G11 and a 7 in G12. Row 11 already has a 7 in it (E11), so we have to place the 7 of the solution for G11 and G12 in G12. This leaves us with only G11 to place the 4.
- We must place a 4 in I10 and a 2 in I12. Row 10 already has a 2 in it (A10) and Row 12 already has a 4 in it (C12), so we have to place the 4 and 2 in the opposite columns of the existing 4 and 2s, and we will end up with a 4 in I10 and a 2 in I12.

After Step 6, you should get this as your final solution:

	A	B	C	D	E	F	G	H	I	J	K	L
1	9	7	5	2	12	8	10	6	3	11	4	1
2	10	5	8	7	4	1	6	3	9	2	11	12
3	6	2	1	3	11	5	8	7	10	9	12	4
4	8	4	10	12	1	3	5	2	7	6	9	11
5	4	11	6	9	5	10	3	1	8	12	7	2
6	12	1	2	5	9	7	11	4	6	3	8	10
7	1	9	3	4	10	2	12	5	11	7	6	8
8	7	8	9	11	3	4	2	10	12	1	5	6
9	11	12	7	10	2	6	9	8	1	5	1	3
10	2	10	11	6	8	12	1	9	4	1	3	7
11	3	6	12	8	7	9	4	11	1	10	2	5
12	11	3	4	1	6	11	7	12	2	8	10	9